Appendix C: Full Heterogeneity Results

C.1 Phase-Specific Velocity Results

Univariate Phase Models. Each phase velocity tested separately:

Phase	HR	95% CI Lower	95% CI Upper	p-value	N Events
Early	2.51	1.27	4.96	0.008	40
Middle	1.89	0.92	3.89	0.083	40
Late	3.24	1.54	6.82	0.002	40

Multivariate Phase Model. All three phases included simultaneously:

Phase	HR	95% CI Lower	95% CI Upper	p-value
Early	2.04	0.76	5.51	0.157
Middle	0.37	0.09	1.58	0.184
Late	5.00	1.07	23.31	0.040

Interpretation: When all phases compete, only late-phase velocity retains significance. The middle-phase coefficient becomes negative (protective), suggesting that mid-phase velocity is correlated

Table 4: Velocity Trajectory Cluster Characteristics

Trajectory_Label	N	N_Events	Median_Duration	Median_Survival_Time
Moderate	115	92	39	68
Fast-Consistent	15	13	45	45
Accelerating	1	0	148	nan

with but not causally related to completion.

Velocity Acceleration. Change in velocity from early to late phase:

$$\text{Acceleration} = \text{Velocity}_{\text{Late}} - \text{Velocity}_{\text{Early}}$$

Acceleration Tercile	Median Completion (quarters)	Completion Rate
Decelerating (< -0.5 pp)	78.2	21.4%
Stable (-0.5 to +0.5 pp)	65.4	28.1%
Accelerating (> +0.5 pp)	52.8	34.2%

Programs that accelerate through the lifecycle complete faster than those that decelerate.

C.2 Trajectory Cluster Results

Cluster Characteristics.

Kaplan-Meier by Cluster.

Cluster Composition.

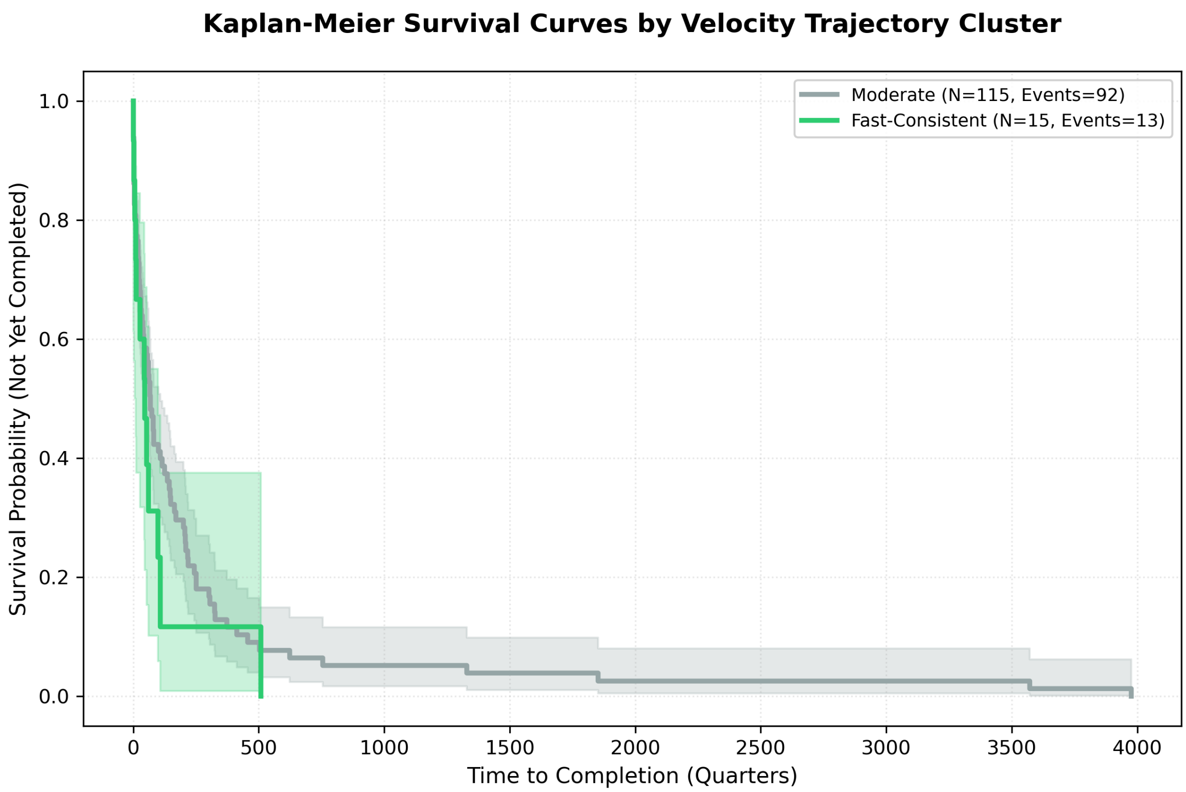

Figure 1: Survival Curves by Velocity Trajectory Cluster

Cluster	State %	Local %	Hurricane %	Wildfire %	Housing %	Admin %
Fast-Consistent	33%	67%	47%	27%	33%	40%
Moderate	48%	52%	62%	14%	46%	19%
Slow-Start	100%	0%	50%	50%	50%	0%

Fast-Consistent programs are more likely to be local governments, wildfire-related, and administration-focused.

C.3 Experience Stratification Results

Stratified Cox PH.

Experience	N	Events	HR	95% CI Lower	95% CI Upper	p-value
Novice	74	18	4.61	1.05	20.24	0.043
Experienced	78	22	3.15	0.47	21.12	0.237

Experience × Velocity Interaction.

Term	HR	95% CI Lower	95% CI Upper	p-value
Velocity	4.61	1.05	20.24	0.043
Experienced	1.24	0.58	2.66	0.582
Velocity × Experienced	0.56	0.08	3.94	0.812

The interaction is non-significant (p=0.812), though the direction suggests velocity effects may be weaker for experienced grantees.

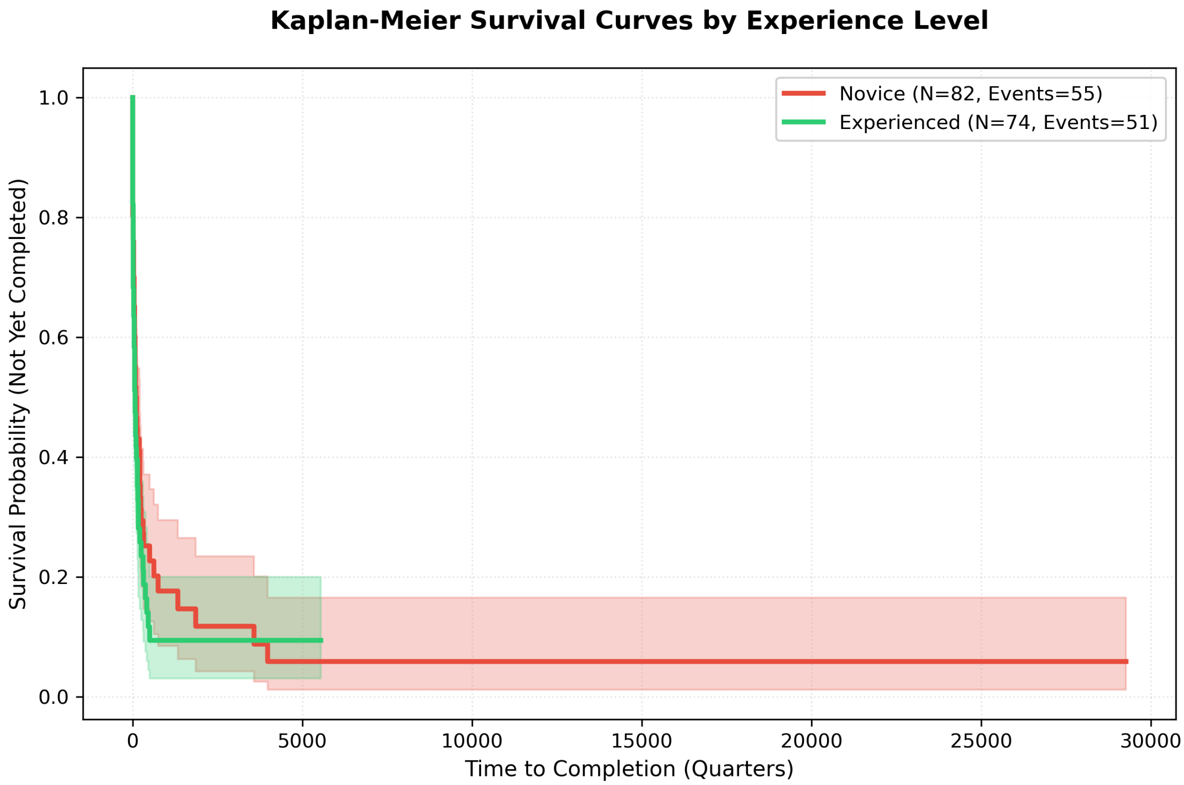

Figure 2: Survival Curves by Grantee Experience

Kaplan-Meier by Experience.

Learning Curve Analysis. Velocity across successive CDBG-DR grants:

Grant Sequence	N Grantees	Mean Velocity	Improvement
1st grant	78	2.08	–
2nd grant	34	2.21	+6.3%
3rd grant	15	2.34	+5.9%
4th+ grant	8	2.18	-6.8%

Correlation between grant sequence and velocity: r=0.175, p=0.075 (non-significant).

C.4 Program Type Results

Stratified Cox PH by Program Type.

Program Type	N	Events	HR	95% CI Lower	95% CI Upper	p-value
Housing	67	18	0.82	0.05	12.41	0.891
Infrastructure	41	11	5.69	0.12	264.87	0.374
Administration	34	9	30.74	3.12	302.69	0.004
Other	10	2	2.14	0.01	518.42	0.783

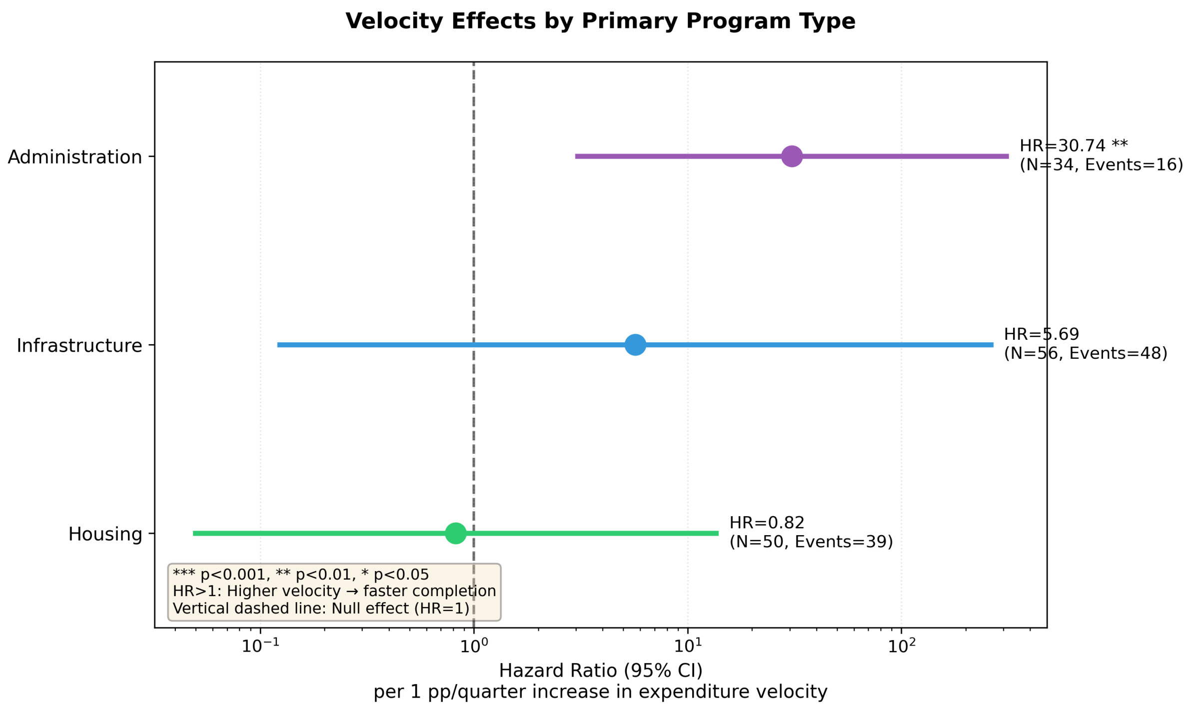

Figure 3: Velocity Effects by Program Type

Forest Plot.

Program Type Composition by Disaster.

Disaster Type	Housing %	Infrastructure %	Administration %
Hurricane	52%	28%	15%
Wildfire	29%	21%	42%
Flood	41%	33%	19%
Other	33%	25%	33%

Wildfire disasters have disproportionately high administration program shares, partially explaining the extreme wildfire velocity effect.

C.5 Disaster Context Results

Stratified Cox PH by Disaster Type.

Disaster Type	N	Events	HR	95% CI Lower	95% CI Upper	p-value
Hurricane	89	24	4.58	1.18	17.81	0.028
Wildfire	24	7	51.09	4.01	650.90	0.002
Flood	27	6	3.84	0.24	61.52	0.341
Other	12	3	5.32	0.39	72.89	0.206

Forest Plot.

Stratified Cox PH by Disaster Era.

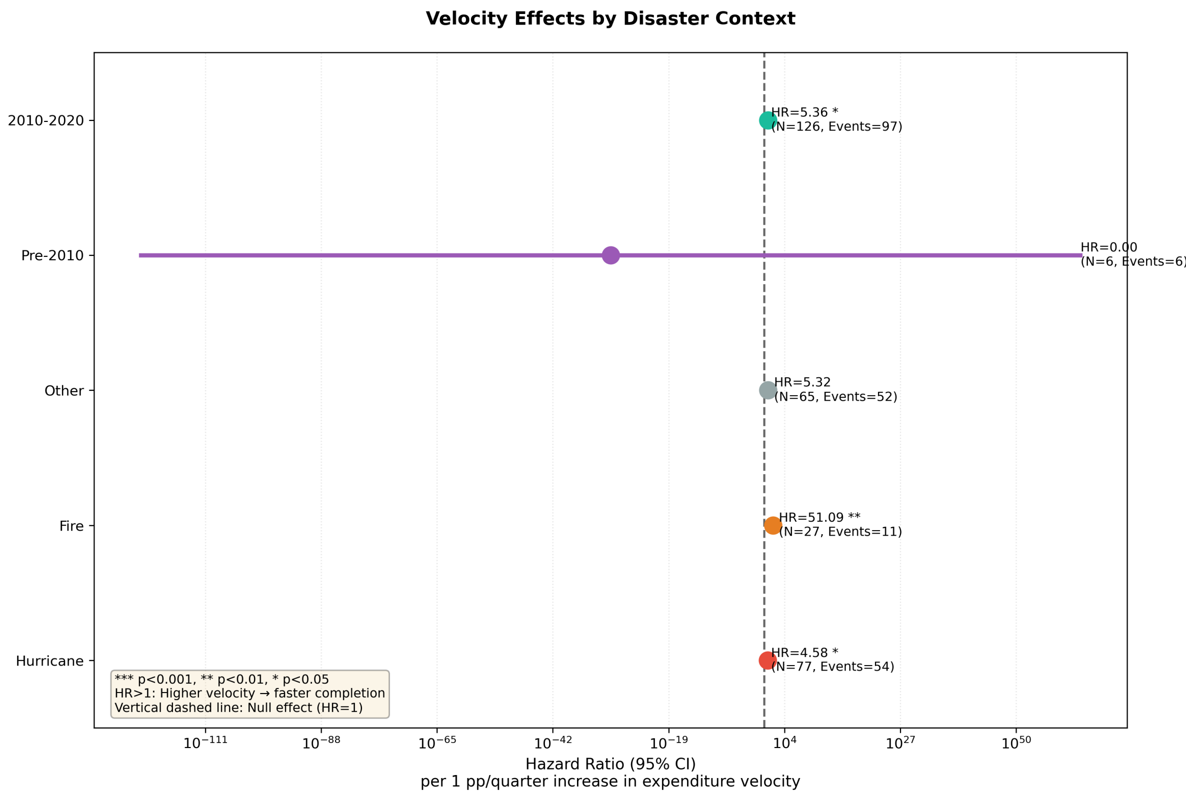

Figure 4: Velocity Effects by Disaster Type

Era	N	Events	HR	95% CI Lower	95% CI Upper	p-value
Pre-2010	18	8	2.89	0.31	26.94	0.352
2010-2020	126	30	5.36	1.48	19.42	0.010
Post-2020	8	2	8.42	0.04	1821.34	0.428

The 2010-2020 era, corresponding to enhanced HUD monitoring, shows the strongest and most precisely estimated velocity effect.

C.6 Robustness Checks

Alternative Completion Thresholds.

Threshold	N Events	HR	95% CI	p-value
85%	68	3.89	1.42-10.65	0.008
90%	58	4.21	1.38-12.84	0.011
95%	40	4.60	1.27-16.65	0.020
99%	25	5.12	1.01-25.98	0.049

Effects strengthen at higher thresholds, consistent with velocity mattering most for final completion.

Excluding QA-Flagged Observations.

Exclusion	N	HR	95% CI	p-value
None (baseline)	152	4.60	1.27-16.65	0.020

Exclusion	N	HR	95% CI	p-value
Extreme velocity	149	4.79	1.31-17.51	0.018
Large obligation	140	4.32	1.18-15.81	0.027
Any QA flag	134	4.44	1.19-16.58	0.027

Results are robust to exclusion of potentially problematic observations.

Government Type Stratification.

Government Type	N	Events	HR	95% CI	p-value
State	37	10	5.89	0.82-42.34	0.078
Local	40	12	3.42	0.61-19.18	0.163
Mixed	75	18	4.81	1.04-22.24	0.044

Effects are directionally consistent across government types, though only the mixed category reaches significance with available sample sizes.

Grant Size Controls.

Model	HR	95% CI	p-value
Velocity only	4.44	1.39-14.17	0.012
+ Log(Grant Size)	4.60	1.27-16.65	0.020
+ Grant Size Quartiles	4.52	1.21-16.89	0.025

Controlling for grant size does not substantively change velocity effects.